Project 3: Multi-Agent Deep Deterministic Policy Gradients in the Overcooked Environment

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1 Introduction

The Overcooked environment presents a challenging multi-agent reinforcement learning (RL) problem where two agents, modeled as chefs in a kitchen, must coordinate to maximize the number of soups delivered within a strict 400-step episode limit. The task requires that each agent perform actions such as collecting onions, preparing dishes, and delivering soup while navigating three different kitchen layouts. The core objective is to ensure efficient, error-free collaboration that achieves the highest possible soup delivery rate. From an RL perspective, the Overcooked environment presents a variety of unique and significant challenges. The 2 agents can move up, down, left, right, stay, and "interact." The chefs must engage in a high level of coordination to avoid missteps that could hinder their efficiency, such as dropping ingredients. Each layout varies in difficulty and requires different levels of cooperation that need clear collaboration between both agents. These factors create a complex decision space, accompanied by the need for communication and credit assignment between the agents (Foerster et al., 2016; Lowe et al., 2017). This multi-agent setting, along with the inherent difficulties of coordinating actions within a limited action space, makes the Overcooked environment a challenging yet exciting scenario for developing advanced RL solutions. Our primary goal for this project is to achieve a mean soup delivery of seven or more across all three layouts (cramped room, asymmetric advantages, and forced coordination) by successfully implementing a Multi-Agent Deep Deterministic Policy Gradient (MADDPG) solution (Lowe et al., 2017).

2 Methods

2.1 Overview of Approach

The Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm was selected for this project due to its suitability for environments that require coordinated agent behavior within a shared, multi-agent setting. MADDPG is particularly effective for the Overcooked environment because it combines centralized training with decentralized execution (Lowe et al., 2017). This architecture enables agents to learn policies that enhance collective performance while operating independently during execution. A key feature of MADDPG is the use of a **centralized critic**, which allows the agents to incorporate the full environment state during training. This centralized perspective provides richer feedback by observing the combined states and actions of all agents, leading to better credit assignment. In the Overcooked environment, where efficient task-sharing and role allocation between the two chefs are essential, the centralized critic allows each agent to learn more effective strategies based on the outcomes of collaborative actions, helping to maximize soup deliveries (Foerster et al., 2018). During execution, MADDPG enables each agent to make decisions based on only its own observations. This **decentralized approach** is critical for real-time multi-agent tasks, where each agent must act independently while remaining responsive to the actions of others (Foerster et al., 2018). In Overcooked, this means that each chef can respond flexibly to the changing layout and environment dynamics while still aligning with the overarching goal of maximizing soup deliveries. The decentralized nature of MADDPG also ensures that each agent's policy generalizes well across different layouts, providing robustness in various kitchen configurations (Lowe et al., 2017).

2.2 Adapting DDPG to MADDPG

This project's implementation of MADDPG builds directly on the DDPG agent developed in Project 2 for the lunar lander environment, with key modifications made to adapt it to a multi-agent setting. Unlike the single-agent configuration of Project 2, the Overcooked environment requires that two agents work collaboratively. Each agent operates with its own actor and critic networks, allowing them to individually assess the impact of their actions while also benefiting from the shared learning that a centralized critic enables (Lowe et al., 2017). To facilitate this joint learning, a shared replay buffer was introduced to store transitions from both agents, enabling them to sample from and learn jointly from past experiences. This shared learning experience allows the agents to adjust their policies to the specific coordination challenges present in the Overcooked environment (Foerster et al., 2016). Additionally, the observation space for each agent was expanded to capture not only its own state but also that of the other agent and critical kitchen elements like dishes and onions. These observations were adapted for the MADDPG architecture, ensuring that each agent's policy is shaped by the comprehensive context of the collaborative environment (Iqbal & Sha, 2019).

2.3 State and Action Space

In the Overcooked environment, each agent has access to a fully observable **state**, represented by a 96-element vector that encodes critical information about the agent's surroundings. This vector includes details on each agent's orientation, object currently held, proximity to items like onions, pots, dishes, and serving areas, as well as information about the presence and status of other agents (Foerster et al., 2016). These state elements enable each agent to make decisions informed by its own position relative to its partner and relevant kitchen objects. Given the collaborative nature of the task, this comprehensive state representation is essential for agents to plan and execute tasks that maximize the number of soups delivered.

The **action space** in Overcooked is discrete and comprised of six possible actions: up, down, left, right, stay, and "interact." The "interact" action is context-sensitive, allowing the agent to engage with nearby items, such as placing onions in a dish or serving soup when positioned next to the serving area. Since MADDPG produces continuous action values, a mapping strategy was implemented to translate these continuous values into one of the six discrete actions required by the Overcooked environment. Specifically, the continuous outputs were mapped to discrete actions based on thresholds that align with Overcooked's grid-based layout (Rashid et al., 2020). This adaptation allows the agents to retain the precision and flexibility of continuous control while conforming to the environment's discrete action requirements.

3 Algorithm Description

The MADDPG algorithm utilized in this project is structured to effectively support each agent's learning and coordination within the Overcooked environment, leveraging actor and critic networks, a multi-agent replay buffer, reward shaping, and parameter updates to drive efficient collaboration and task execution (Foerster et al., 2016; Lowe et al., 2017).

Each **agent's architecture** comprises actor and critic networks with two fully connected hidden layers. The hidden layers are equipped with batch normalization, which standardizes the activations to stabilize learning and help prevent overfitting. To optimize training, a learning rate schedule is employed, where the learning rate decays gradually over training episodes (Iqbal & Sha, 2019). This adaptive adjustment helps the agents refine their policies in later stages of training when precision in action selection becomes more crucial for maximizing rewards. Each agent's actor network outputs a continuous action that is subsequently mapped to discrete values required by the environment, while the critic network evaluates the effectiveness of the actions in the context of the collaborative environment.

The **replay buffer** is a shared memory structure that accommodates multi-agent experiences, storing observations, actions, rewards, and terminal states for each agent. This buffer enables the agents to sample from a broader distribution of experiences, enhancing their learning by exposing them to a diverse range of scenarios (Iqbal & Sha, 2019). Each transition includes not only the individual agent's observations and actions but also the combined states and actions of both agents. This centralized storage enables a more holistic training process, where each agent's experience is evaluated within the full context of the environment, helping refine its policy with regard to the other agent's behavior.

Reward shaping is crucial to encourage collaboration between the agents. In this implementation, rewards are assigned for critical tasks that directly contribute to the goal of delivering soups, such as placing onions in pots, picking up dishes, and delivering finished soups (Foerster et al., 2018). This reward structure is designed to reinforce positive interactions and promote teamwork, ensuring that both agents contribute meaningfully to the overall objective. By distributing rewards for intermediate tasks, agents are incentivized to perform complementary actions and support each other's goals within the limited 400-step episode.

To **maintain training stability**, parameter updates to the actor and critic networks employ a soft-update mechanism with a small tau value. This approach gradually updates the target networks with a fraction of the current networks' weights at each step, allowing the agents to adapt their policies incrementally (Rashid et al., 2020). This technique prevents large, abrupt shifts in policy, ensuring smoother convergence and reducing the risk of destabilizing the learning process. The actor and critic networks are updated by minimizing the loss between predicted and actual values in the critic network, with the actor network aiming to maximize the expected return as evaluated by the critic. Together, these components form a cohesive learning framework that supports each agent's ability to adapt and improve, resulting in efficient multi-agent collaboration and an increased rate of successful soup deliveries across varied kitchen layouts.

4 Experimental Setup

4.1 Environment Setup

The Overcooked environment was configured with three distinct layouts, **cramped room, asymmetric advantages, and forced coordination**, each presenting unique challenges that are valuable for evaluating the effectiveness of multi-agent collaboration (Carroll et al., 2019). The cramped room layout is compact, requiring precise agent movements to avoid collisions and bottlenecks in the limited space. This layout tests each agent's ability to coordinate actions in a constrained environment where inefficiencies or errors can quickly disrupt the workflow. The asymmetric advantages layout introduces an imbalanced configuration where each agent has easier access to certain resources. This layout evaluates the agents' ability to exploit their unique positional advantages and effectively distribute tasks to maximize overall productivity. Finally, the forced coordination layout requires high levels of collaboration, as agents must actively coordinate their actions to navigate the environment and reach task-critical locations. In this layout, optimal performance hinges on each agent's capacity to anticipate and adapt to the other's actions, making it an ideal setting for testing the effectiveness of collaborative strategies.

4.2 Training Protocol

In terms of **training protocol**, each episode was limited to a fixed 400-step horizon, with training conducted over **60,000** episodes to ensure policy convergence across all layouts. The hyperparameters, including learning rate, gamma, batch size, and tau for soft updates, were selected based on initial exploratory experiments and kept consistent across all layouts. This approach ensures that the trained agents rely on generalizable strategies rather than layout-specific optimizations. The choice to maintain fixed hyperparameters across environments allows for a robust evaluation of the agents' adaptability and ensures that the learned policies can be applied to different layouts without requiring custom tuning. Additionally, this protocol supports the overarching goal of developing agents capable of general multi-agent collaboration across diverse configurations, providing insights into the generalizability of the MADDPG approach (Lowe et al., 2017) in complex, multi-agent environments. This experimental setup offers a comprehensive assessment of the MADDPG model's robustness, adaptability, and collaboration efficacy within the Overcooked environment by training and evaluating the agents across these three contrasting layouts with consistent hyperparameters (Foerster et al., 2018).

5 Results and Analysis

5.1 Performance on Layouts

Each layout posed distinct coordination challenges that tested the agents' ability to adapt their collaborative strategies within the Overcooked environment (Carroll et al., 2019). Over the course of training, we tracked the **number of dish and onion pickups per 1000 episodes**, which serve as proxies for the agents' interaction with critical resources in the environment. These pickups are essential actions that directly contribute to the agents' primary objective of delivering soups efficiently.

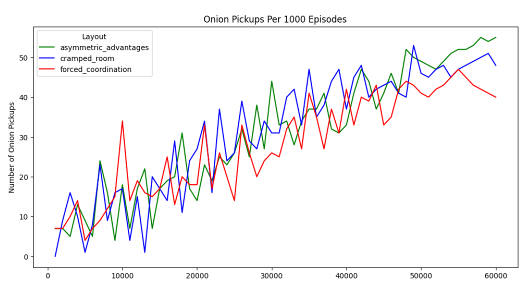

Figure 1: *Onion pickups over training episodes*

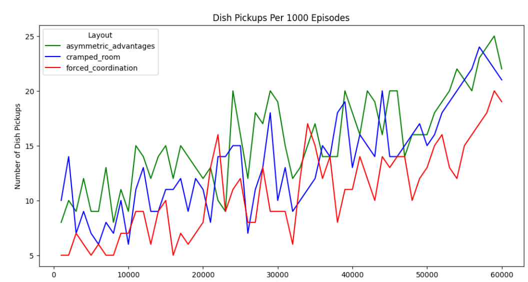

Figure 2: *Dish pickups over training episodes*

The plots in Figures 1 and 2 illustrate the agents' learning curves in terms of dish and onion pickups across training episodes. Notably, the trend lines reveal increasing pickups over time, suggesting that the agents gradually learned to prioritize essential tasks and optimize resource collection using MADDPG (Lowe et al., 2017).. In both metrics, the "asymmetric advantages" and "cramped room" layouts show relatively higher and more consistent pickups compared to the "forced coordination" layout, indicating smoother collaboration in simpler layouts.

For a more granular look at the agents' consistency and stability, these metrics allow us to infer the agents' capacity to maintain reliable task execution within each layout. In the "asymmetric advantages" and "cramped room" layouts, the agents display relatively steady growth in both dish and onion pickups, reflecting a well-coordinated strategy that effectively meets task requirements. However, in the "forced coordination" layout, the agents show more variability in their pickups, particularly with dish pickups. This increased variability suggests that the forced coordination layout demands more precise timing and complex role-sharing, leading to challenges in achieving consistent performance.

5.2 Multi-Agent Metrics

To gain a deeper understanding of the agents' behavior and the effectiveness of the MADDPG approach (Lowe et al., 2017), we tracked two key performance metrics: **average delivered soups**

per 1000 episodes over time and **delivered soups per 1000 episodes (evaluation)**. These metrics allow us to assess the agents' efficiency in fulfilling the primary objective, which is delivering soups consistently across different layouts.

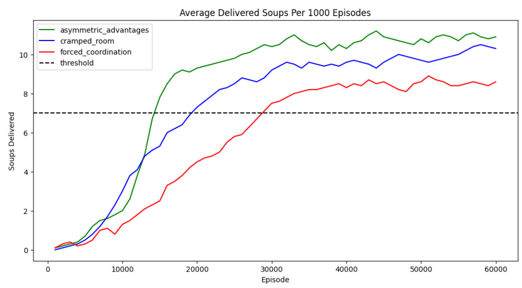

Figure 3: *Delivered soups over training episodes*

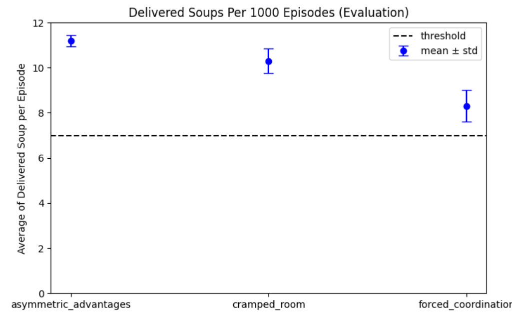

Figure 4: *Evaluation of average soups delivered per episode*

The **average delivered soups per episode** (shown in the line chart) provides insight into how well the agents improve over time, gradually approaching and surpassing the project's threshold of seven soups per episode. This trend is observed across all three layouts—*asymmetric advantages*, *cramped room*, and *forced coordination*. Notably, the agents achieve a steady and rapid increase in the *asymmetric advantages* layout, reaching just over 11 soups per episode, which indicates strong coordination and efficient task execution. In contrast, the *forced coordination* layout shows a slower increase and more gradual progression, reflecting the additional challenge of coordinating tasks in a more complex environment (Foerster et al., 2018). This layout's performance, while improved, does not reach the efficiency seen in the other two layouts, highlighting the need for advanced coordination strategies when spatial separation is enforced.

The **delivered soups per 1000 episodes (evaluation)** (shown in the bar chart) summarizes the agents' performance after training by presenting the average number of soups delivered with error bars representing standard deviation. The results show that the agents met the project's target of at least seven soups per episode across all layouts. However, there is a notable difference in mean performance and consistency. In the *asymmetric advantages* and *cramped room* layouts, the agents delivered over 10 soups per episode, with low variability, indicating consistent, reliable performance. Conversely, the *forced coordination* layout exhibits a lower average and greater variability, suggesting that maintaining coordination in this environment is more challenging and prone to fluctuations in performance (Carroll et al., 2019).

These results highlight the adaptability of the MADDPG approach across varying levels of task complexity. The agents learn to maximize efficiency in simpler layouts while displaying moderate success in more complex configurations. The performance variations across layouts underscore the importance of environment-specific strategies, particularly in setups like the *forced coordination* layout, where optimal task allocation and spatial awareness are essential for achieving consistent outcomes.

5.3 Qualitative Observations

In addition to quantitative metrics, qualitative observations of the agents' behaviors offered valuable insights into the effectiveness of the MADDPG approach (Lowe et al., 2017). As training progressed, the agents began to autonomously divide tasks in a way that optimized the workflow within each layout. For instance, in several episodes, one agent consistently prioritized gathering ingredients, transporting onions to the pot, and managing the initial preparation, while the other agent assumed responsibility for monitoring cooking times, plating soups, and delivering finished dishes. This division of labor emerged naturally through training and was particularly effective in the cramped room and asymmetric advantages layouts, where task specialization minimized collisions and bottlenecks (Foerster et al., 2018).

In the forced coordination layout, the agents exhibited a more dynamic role-switching behavior, with each adapting to the other's movements as they navigated to opposite ends of the kitchen. This adaptability is indicative of a learned understanding of each other's roles, which is reinforced by the centralized training in MADDPG (Carroll et al., 2019). These observations suggest that the agents successfully learned role specialization and adaptive collaboration, underscoring MADDPG's capacity for fostering emergent cooperative behaviors in complex, multi-agent environments.

6 Discussion

6.1 Effectiveness of MADDPG and Reward Shaping

The MADDPG algorithm proved highly effective in fostering coordinated behavior between the two agents in the three Overcooked environment layouts (Lowe et al., 2017). Through its centralized training approach, MADDPG enabled the agents to learn policies that accounted not only for their individual actions but also for the actions of their teammate, a feature that is crucial in complex, multi-agent settings like Overcooked. The centralized critic provided a fuller picture of the environment by observing the complete state and action pairs, which allowed for more effective credit assignment and a better understanding of interdependent actions (Foerster et al., 2018). This enhanced the agents' ability to coordinate, especially in layouts requiring close interaction or well-timed task handoffs (Carroll et al., 2019).

Reward shaping played an instrumental role in guiding the agents toward optimal behavior by assigning incremental rewards for actions that supported the main objective of delivering soups (Foerster et al., 2016). By emphasizing tasks like placing ingredients in pots or delivering finished dishes, the reward shaping encouraged agents to focus on productive actions and reinforced behaviors aligned with efficient task completion. This incentive structure promoted teamwork, as the agents learned to allocate tasks and operate cohesively, avoiding unnecessary movements that could disrupt workflow. However, there were some limitations to the MADDPG setup, including slower convergence in certain layouts. The forced coordination layout, in particular, posed stability challenges, as the agents sometimes struggled to maintain consistent task allocation. This highlights a known issue in MADDPG, where achieving stable policies can be more complex in scenarios with high interdependence between agents.

6.2 Comparison with Project 2 Solution

Compared to the single-agent DDPG solution implemented in Project 2 for the lunar lander environment, the MADDPG approach introduced here demonstrates marked improvements in multi-agent coordination and adaptability (Lowe et al., 2017). Unlike DDPG, which was developed for a single-agent setting, MADDPG is specifically designed for collaborative environments, allowing the agents to learn from their own actions and their teammate's actions. This collaborative learning resulted in significant performance gains, particularly in the asymmetric advantages and forced coordination layouts, where explicit coordination is essential for optimal performance. The MADDPG agents exhibited a robustness across layouts that were absent in the Project 2 DDPG setup, showing a capacity to adapt to various environmental challenges while maintaining high efficiency in task execution.

6.3 Challenges

While we selected and kept consistent hyperparameters across layouts to ensure robustness and test the adaptability of the MADDPG approach, this choice came with certain trade-offs. Fixed hyperparameters supported policy generalization, but achieving an ideal balance between exploration and coordination proved challenging (Foerster et al., 2016). Certain layouts presented unique demands that may have benefited from slight adjustments in hyperparameters, particularly regarding exploration rates and task prioritization, to improve convergence rates. These observations highlight that although fixed hyperparameters serve the overarching goal of evaluating robustness, sensitivity to specific hyperparameter values remains a challenge in multi-agent settings, where each environment can present distinct coordination complexities.

Reward shaping also presented some challenges, as specific shaping choices had a significant impact on agent behaviors. For instance, overly emphasizing rewards for item placement sometimes led to suboptimal task division between agents, as they prioritized certain actions over others (Foerster et al., 2018). This required careful calibration to promote a balanced strategy without destabilizing learning. Additionally, while MADDPG's centralized critic enabled more effective credit assignment, the complex interactions in certain layouts led to non-trivial coordination issues (Carroll et al., 2019). In some cases, agents struggled to effectively split roles, which impacted their ability to deliver soups efficiently in environments with tighter spaces or more task dependencies.

Another challenge was achieving a balance between exploration and exploitation. Fixed exploration rates across layouts sometimes hindered effective exploration in more complex layouts, potentially limiting the agents' ability to discover optimal strategies (Foerster et al., 2016). Lastly, the multi-agent setup increased computational demands, with slower convergence observed in layouts requiring more intricate coordination. These computational requirements highlighted the potential limitations of MADDPG in highly interactive multi-agent environments, where complex coordination is essential for success.

7 Conclusion

This project demonstrated the effectiveness of the Multi-Agent Deep Deterministic Policy Gradient (MADDPG) algorithm in addressing the challenges of multi-agent coordination within the Overcooked environment (Lowe et al., 2017; Carroll et al., 2019). By implementing MADDPG with a centralized critic and decentralized execution, we developed a framework that supports cooperative strategies and enables each agent to act independently while optimizing for shared objectives. Additionally, the use of tailored reward shaping proved critical for encouraging behaviors that emphasize teamwork and efficient task-sharing, highlighting the importance of strategic reward design in multi-agent reinforcement learning (RL) (Foerster et al., 2016). Together, these components facilitated a strong approach to handling the unique demands of the Overcooked environment.

The findings of this project underscore MADDPG's capability to enable collaboration across diverse layouts, achieving the target goal of seven or more soup deliveries per episode across each environment layout. This represents a significant improvement over single-agent methods, which are often limited by their inability to effectively coordinate in complex, task-oriented environments. The project results also provided insights into how fixed hyperparameters influence generalizability, as well as the challenges posed by multi-agent credit assignment and exploration strategies (Foerster et al., 2018). These observations contribute to a broader understanding of the conditions under which MADDPG excels and offer valuable guidance for future applications of multi-agent RL in similarly structured environments.

Although this was one of the most challenging projects I ever had to complete, it was among one of the most rewarding ones as well. It provided a comprehensive exploration of multi-agent RL challenges, from reward shaping to architecture selection and training stability. Working within the Overcooked environment reinforced the practical applications of RL in real-world multi-agent problems, where agents must adapt and collaborate to achieve complex objectives (Carroll et al., 2019). Through this project, I have gained a deeper appreciation for the intricacies of multi-agent systems and the potential of reinforcement learning to solve collaborative, high-stakes problems, both within controlled environments like Overcooked and in broader, real-world contexts.

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